nature mental health

Analysis

https://doi.org/10.1038/s44220-026-00667-9

Vertex-wise cortical abnormalities in major depressive disorder from 64 cohorts from the DIRECT and ENIGMA MDD consortia

Received: 24 July 2025

Accepted: 8 May 2026

Published online: 15 June 2026

Check for updates

A list of authors and their affiliations appears at the end of the paper

Major depressive disorder (MDD) is common and disabling, yet reported brain structural differences vary across studies. Here we performed a large vertex-wise (point-by-point) meta-analysis of cortical thickness and surface area using harmonized magnetic resonance imaging processing across 64 cohorts from the Enhancing NeuroImaging Genetics through Meta-Analysis (ENIGMA) MDD and Depression Imaging Research Consortium (DIRECT) consortia (5,736 patients; 6,538 controls). We show significantly lower cortical thickness in patients with MDD in multiple brain regions, including the inferior parietal, lateral occipital, superior parietal, medial and lateral orbitofrontal, anterior and posterior cingulate, and precentral gyri, with cortical surface area showing no significant differences. Effects were most pronounced in adults with acute depression, whereas adolescents showed no significant case–control differences. Antidepressant medication use at scanning was associated with more extensive thinning, although effect sizes remained modest (mostly |Cohen's d| < 0.20). This high-resolution, globally generalizable map can support studies of mechanisms and help evaluate structural markers of the clinical course and treatment response.

Major depressive disorder (MDD) is one of the most prevalent mental disorders. Approximately 350 million people suffer from depression each year across the globe[1]. Effective treatment is critical to reduce the tremendous burden of MDD, yet first-line treatments are only moderately effective, with response rates of under 50% for both psychotherapy[2] and antidepressant monotherapy[3]. Neuroscientific research aims to identify biomarkers and develop better treatments through an improved understanding of the neural and genetic basis of MDD. Large-scale studies have identified structural brain alterations associated with MDD, including lower cortical thickness and gray matter (GM) volumes, predominantly in inferior/superior/middle frontal, orbitofrontal, anterior and posterior cingulate, temporal and insular regions, as well as the hippocampus[4–9]. These alterations were most pronounced in individuals experiencing an acute episode of MDD, with an earlier age of onset and taking antidepressants at time of scanning. Findings on cortical surface area differences have been more inconsistent. For example, our previous Enhancing NeuroImaging Genetics through Meta-Analysis (ENIGMA) MDD study showed lower surface area across a number of regions only in adolescents with MDD (aged <21 years)[7], whereas a similar multicenter meta-analysis from the Japanese Cognitive Genetics Collaborative Research Organization (COCORO) consortium observed surface area alterations in adults with MDD (aged >21 years)[5].

Overall, in large-scale, single-center (for example, using the Nathan Kline Institute-Rockland Sample) or coordinated multicenter studies (for example, from the ENIGMA MDD consortium, COCORO consortium and UK Biobank), the effects of case–control differences in MDD have been found to be relatively small (Cohen's d effect sizes ranging from 0.03 to 0.37)[5–7,9]. However, many of these studies included relatively coarse measures of cortical regions (for example, derived from the Desikan–Killiany atlas in FreeSurfer)[10], averaging across many voxels and functional subregions that may be more or less affected in MDD. Voxel- or vertex-level analyses, which provide a finer spatial resolution, may unveil more localized effects that might be diluted in broader region-based averages and may yield larger effect sizes. For example, in our previous ENIGMA MDD vertex-level analysis of

A full list of affiliations appears at the end of the paper. e-mail: yancg@tsinghua.edu.cn; lianne.schmaal@unimelb.edu.au

 https://doi.org/10.1038/s44220-026-00667-9

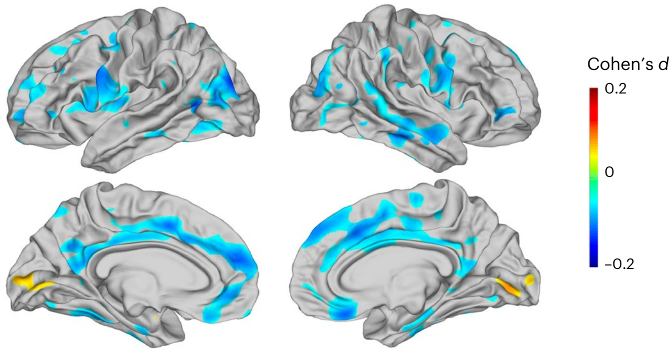

Fig. 1 | Cortical thickness differences between all patients with MDD and HCs (5,256 MDDs versus 6,320 HCs). The values represent Cohen's *d* (positive values indicate MDD > controls).

subcortical volumes, we identified alterations in both the hippocampus and amygdala[11], while we only observed lower hippocampal volumes in our analysis of atlas-derived regional measures[9].

Here, within an individual participant data-based meta-analytic framework, we pooled thickness and surface area measures from every vertex of the cortex from data from the ENIGMA MDD and the Depression Imaging Research Consortium (DIRECT) consortia[12]. ENIGMA MDD is a global initiative that pools neuroimaging, demographic, genetic and clinical data from existing populations around the world to identify brain alterations in MDD and associated phenotypes. The DIRECT consortium similarly pools existing neuroimaging, demographic and clinical data from China. By combining data from these two consortia, we not only created the largest cohort so far of 5,736 patients with MDD and 6,538 healthy controls (HCs) to identify localized cortical thickness and surface area alterations in MDD, but also increased diversity, thereby providing results that are robust and replicable across many different geographic locations and cultures worldwide. We also examined whether specific subgroups of individuals with MDD (that is, based on age, age of onset, sex, remission status, recurrence status and use of antidepressant medication) show more pronounced cortical thickness and surface area differences.

All patients with MDD versus controls

The DIRECT consortium (23 Chinese cohorts) and the ENIGMA MDD consortium (41 international cohorts) analyzed FreeSurfer vertex data from 12,274 participants (5,736 patients with MDD and 6,538 HCs; Supplementary Tables 1 and 2). Meta-analyses of cortical thickness and surface area abnormalities were performed on 11,576 individuals (5,256 MDDs versus 6,320 HCs) after quality control. All models were adjusted for age and sex. In the analysis for surface area, the estimated total intracranial volume was also used as a covariate, while no adjustment was performed for cortical thickness. The results revealed that patients with MDD had significantly thinner cortex in several regions, with the largest effect size (Cohen's $d = -0.14$) in the left inferior parietal, lateral occipital and superior parietal cortices. Conventionally, Cohen's *d* values of around 0.2, 0.5 and 0.8 indicate small, medium and large effects, respectively. In large-scale multisite datasets, increased population heterogeneity inflates overall variance and thus leads to smaller standardized effect sizes. Significantly lower cortical thickness was also observed in the medial and lateral orbital frontal cortices, caudal and rostral anterior cingulate cortices, posterior and isthmus cingulate cortices, inferior frontal gyri, middle and superior frontal gyri, precentral and postcentral gyri, cuneus and precuneus, fusiform gyrus, lingual gyrus, parahippocampal gyrus, supramarginal gyrus, superior, middle and inferior temporal gyri, and lateral occipital cortex. Patients with MDD demonstrated greater thickness compared to HCs in a cluster including the right lingual and pericalcarine gyri (Cohen's $d = 0.09$). Significantly greater thickness was also observed in the left pericalcarine gyrus, lingual gyrus and cuneus (Fig. 1, forest plots in Supplementary Fig. 1 and detailed numbers in Supplementary Table 6). When analyzed separately within each consortium, DIRECT (Supplementary Fig. 12a) showed widespread cortical thinning mainly in the bilateral superior and middle frontal, precentral, inferior parietal and supramarginal regions, extending to the precuneus and posterior cingulate cortices. Meanwhile, ENIGMA (Supplementary Fig. 12b) showed more restricted thinning in the medial orbitofrontal, superior frontal and posterior cingulate areas, accompanied by slight thickening in the occipital cortex. The combined meta-analysis largely reflected the convergence of these two patterns, retaining cortical thinning in the superior frontal cortex and posterior cingulate areas that was consistently observed across both consortia, whereas the more extensive fronto-parietal thinning in DIRECT and the occipital thickening in ENIGMA were attenuated after pooling. We found no differences in cortical surface area, or sex-by-diagnosis or age-by-diagnosis interactions.

Adult patients with MDD

Adult patients with MDD versus controls

When comparing 4,139 adult patients with MDD and 5,190 adult HCs, our findings were similar to the analyses performed on the entire cohort (Fig. 2a). We observed significantly lower cortical thickness in several regions, with the largest effect (Cohen's $d = -0.15$) in a cluster including left inferior parietal, lateral occipital and inferior temporal areas. Significantly lower thickness was also observed in the left precentral, postcentral and middle temporal gyri, and right superior frontal, posterior cingulate and precuneus cortices. Patients with MDD also demonstrated greater thickness in several regions, with the largest effect (Cohen's $d = 0.09$) capturing the left pericalcarine gyrus, lingual gyrus and cuneus. Greater thickness was observed in the left postcentral and right lingual, pericalcarine gyri (Fig. 2a, Supplementary Fig. 2 and Supplementary Table 7). No significant findings were detected for cortical surface area.

The interaction of diagnosis with age or sex in adults

In an interaction analysis of diagnosis and age (4,139 patients with MDD and 5,190 HCs), we observed an interaction between the MDD diagnosis and age on cortical thickness, with the largest effect (Cohen's $d = 0.12$) at the intersection of the left superior parietal, lateral occipital and inferior parietal areas (Fig. 2b, Supplementary Fig. 3 and Supplementary Table 8). In this region, older age was associated with a weaker effect size of MDD-related cortical thickness reduction, indicating that the differences in cortical thickness between MDD and HC may become less pronounced with older age. No significant findings were detected for cortical surface area. We did not find any significant interaction between diagnosis and sex for cortical thickness or for surface area.

The effect of age of onset in adult MDD

When comparing 2,752 patients with adult-onset (>21 years of age) MDD with 4,938 adult HCs, we observed significantly lower cortical thickness in several regions, with the largest effect (Cohen's $d = -0.15$) capturing left precentral and postcentral areas. Significantly lower thickness was also observed in the left insula, inferior frontal gyrus pars opercularis, and lateral orbitofrontal, right inferior and superior parietal, and lateral occipital cortices. Patients with MDD also demonstrated greater thickness, with the largest effect (Cohen's $d = 0.10$) capturing the left pericalcarine gyrus, lingual gyrus and cuneus. Greater thickness was observed in the right lingual and pericalcarine gyri (Fig. 2c, Supplementary Fig. 4 and Supplementary Table 9). When comparing 2,254 adult patients with adult-onset MDD with 844 adult patients with adolescent-onset MDD, and comparing 876 adult patients with an adolescent-onset MDD with 4,349 HCs, we found no differences for cortical thickness or for surface area.

The effect of stage of MDD illness in adult MDD

When comparing patients with adult first-episode MDD with HCs (1,387 MDDs versus 4,398 HCs), we observed lower cortical thickness

 https://doi.org/10.1038/s44220-026-00667-9

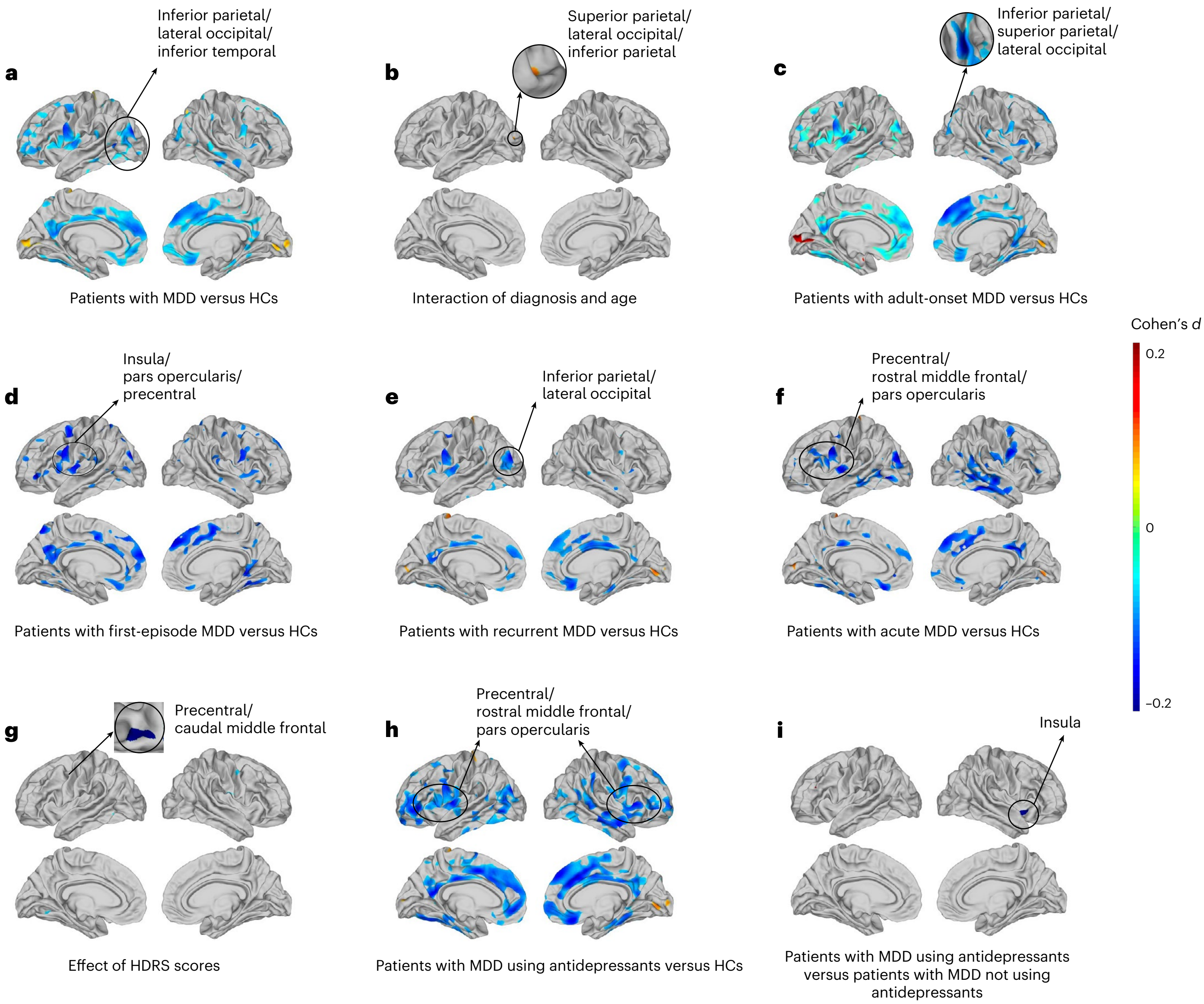

Fig. 2 | Adult (age >21 years) patients' cortical thickness differences in Cohen's *d* with meta-analysis. **a**, Adult patients with MDD and controls (4,139 patients with MDD versus 5,190 HCs). **b**, The interaction of diagnosis and age in adults (4,139 patients with MDD versus 5,190 HCs). **c**, Patients with adult-onset MDD compared to HCs (2,752 patients with MDD versus 4,938 HCs). **d**, Patients with adult first-episode MDD compared to HCs (1,387 patients with MDD versus 4,398 HCs). **e**, Adult patients with recurrent MDD compared to HCs (2,273 patients with MDD versus 4,796 HCs). **f**, Adult patients with acute MDD and controls (1,824 patients with MDD versus 2,360 HCs). **g**, The effect of HDRS scores on adult cortical thickness (on 3,217 patients with MDD). **h**, Adult patients with MDD who were taking antidepressants compared to healthy adult controls (2,172 patients with MDD using antidepressants versus 4,564 HCs). **i**, Adult patients with MDD using antidepressants compared to patients with MDD not using antidepressants (1,724 patients with MDD using antidepressants versus 1,220 patients with MDD not using antidepressants). All results were FDR-corrected to $q < 0.05$ across all cortical vertices in the fsaverage space (left and right hemispheres were combined to $n = 299{,}881$ vertices). All labeled brain regions in the figures are the regions with the strongest effects.

in several regions, with the largest effect (Cohen's $d = -0.19$) capturing the left insula, inferior frontal gyrus pars opercularis and precentral gyrus. Lower thickness was also observed in the left inferior parietal cortex, caudal middle frontal gyrus, and right superior frontal, fusiform and inferior temporal gyri (Fig. 2d, Supplementary Fig. 5 and Supplementary Table 10). When comparing adult patients with recurrent MDD with HCs (2,273 MDDs versus 4,796 HCs), we found lower cortical thickness in many regions, with the largest effect (Cohen's $d = -0.17$) in the left inferior parietal and lateral occipital areas. Significantly lower cortical thickness was also observed in the left precentral and postcentral gyri, left superior parietal cortex, and right medial orbitofrontal and rostral anterior cingulate cortices. We also found greater cortical thickness with the largest effect (Cohen's $d = 0.11$) in the left postcentral area. Greater cortical thickness was observed in the left and right pericalcarine and lingual gyri (Fig. 2e, Supplementary Fig. 6 and Supplementary Table 11).

When we directly compared 1,399 patients with adult first-episode MDD to 2,039 patients with recurrent MDD, we did not find any significant results for cortical thickness. No significant findings were detected for cortical surface area.

The effect of remission in adult MDD

When comparing 1,824 adult patients with acute MDD with 2,360 healthy adult controls, we found significantly lower cortical thickness in many regions, with the largest effect (Cohen's $d = -0.19$) in a cluster including the left precentral, rostral middle frontal and inferior frontal pars opercularis gyri (Fig. 2f). Significantly lower cortical thickness was also observed in the left inferior parietal, lateral occipital, caudal

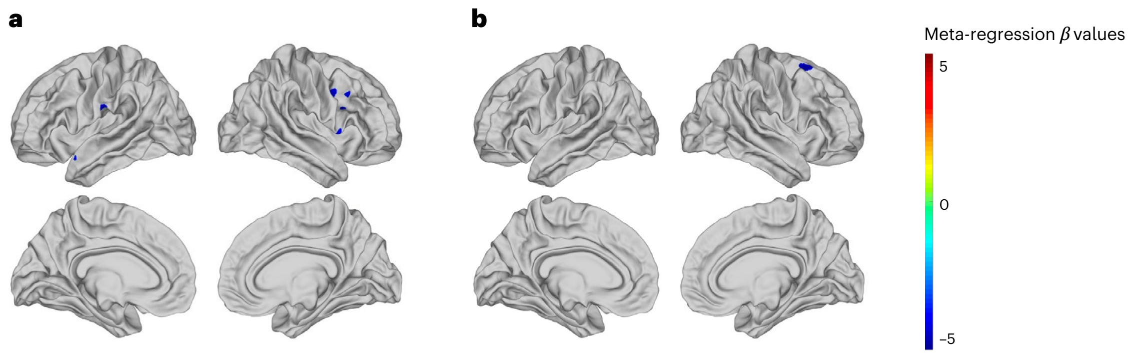

Fig. 3 | Moderating effects. a, Meta-regression effect of FreeSurfer version on cortical thickness. **b**, Meta-regression effect of field strength on cortical thickness. All results were FDR-corrected to $q < 0.05$ across all cortical vertices in the fsaverage space (left and right hemispheres were combined to $n = 299{,}881$ vertices).

middle frontal and superior frontal areas, and right isthmus cingulate, precuneus and posterior cingulate cortices. We also found significantly greater cortical thickness, with the largest effect (Cohen's $d = 0.12$) in the left postcentral area. Greater cortical thickness was observed in the left pericalcarine gyrus and cuneus (as depicted in Supplementary Fig. 7 and Supplementary Table 12). When we compared patients with adult remitted MDD to HCs (276 patients with remitted MDD versus 1,473 HCs), as well as to adult patients with acute MDD (293 patients with remitted MDD versus 802 patients with acute MDD), we did not find any significant differences in cortical thickness. No significant findings were detected for cortical surface area.

The effect of illness severity in adult MDD

In 3,217 adult patients with MDD, there was a significant negative correlation between Hamilton Depression Rating Scale (HDRS) scores and cortical thickness in several regions, with the largest effect (Cohen's $d = -0.09$) in a cluster including the left precentral and caudal middle frontal gyri (Fig. 2g). Significant negative correlations were also observed in the left superior temporal, transverse temporal, and right postcentral cortices (Supplementary Fig. 8 and Supplementary Table 13). That is, the more severe the depression, the greater the reductions in cortical thickness in these areas. No significant findings were detected for cortical surface area.

The associations with antidepressant use in adult MDD

When we compared 2,172 adult patients with MDD who were taking antidepressants at the time of scanning to 4,564 adult HCs, our analysis revealed significantly lower cortical thickness in widespread regions, with the largest effect (Cohen's $d = -0.17$) in a cluster spanning the left insula, inferior frontal gyrus pars opercularis, and precentral gyrus (Fig. 2h, Supplementary Fig. 9 and Supplementary Table 14). Lower cortical thickness was also observed in the left precuneus, posterior cingulate and isthmus cingulate cortex, and right insula, precentral gyrus, and inferior frontal gyrus pars opercularis. We also found significantly greater cortical thickness, with the largest effect (Cohen's $d = 0.11$) in the right lingual and pericalcarine gyri. Greater cortical thickness was also observed in the left postcentral area.

In the direct comparison between 1,724 patients with MDD taking antidepressants and 1,220 patients with MDD not taking antidepressants (excluding cohorts that did not contain both groups of participants), only the right insula (Cohen's $d = -0.24$) showed significantly lower cortical thickness in those using antidepressants (Fig. 2i, Supplementary Fig. 10 and Supplementary Table 15). When comparing patients with MDD not taking antidepressants and HCs (1,188 patients with MDD versus 4,140 HCs), we did not find any significant differences in cortical thickness. No significant findings were detected for cortical surface area.

Cortical alterations in adolescent MDD

Our analysis in adolescents (age <21 years) with MDD ($n = 908$) versus adolescent controls ($n = 883$) did not reveal any significant differences in either cortical thickness or surface area. We repeated all the analyses for adult MDD reported above for adolescent patients with MDD, and none of them revealed significant results. Nonetheless, results for the adolescent group before false discovery rate (FDR) correction show similar or even greater effect sizes for cortical thickness alterations compared to the results in adults (Supplementary Fig. 11).

Moderating effects on cortical thickness and surface area

To investigate whether site-related factors may have influenced the observed effect sizes for the MDD versus HC comparison (across all adult and adolescent patients), we used meta-regression analyses. Scanner vendor and type of cohort (general population versus clinical) did not explain significant variance. As shown in Fig. 3a, when considering FreeSurfer version 6 compared to other versions (mostly version 5 with one site exception of version 7), a significant effect of FreeSurfer version was found: version 6 demonstrated a stronger effect size of MDD-related cortical thickness reduction in the left postcentral and supramarginal gyri, and right caudal middle frontal and precentral gyri (lower in patients with MDD). As shown in Fig. 3b, when considering field strength (1.5 T versus 3 T), higher field strength (3 T) was associated with a stronger effect size for MDD-related cortical thickness reduction in the right superior frontal cortex.

As shown in Supplementary Fig. 13, when considering the proportion of white participants across sites, higher proportions were associated with greater cortical thickness in the left paracentral and precentral regions and the right inferior parietal cortex. We also conducted analyses on the proportions of Asian and African participants, but no significant results were found.

Discussion

In this extensive meta-analytic study, the largest of its kind, we pooled data from the ENIGMA MDD and DIRECT consortia, encompassing 5,736 individuals with MDD and 6,538 HCs. Our findings revealed widespread lower cortical thickness in patients with MDD versus controls across various brain regions, including the prefrontal, temporal, parietal and occipital lobes. The most significant difference was found in a cluster spanning the inferior parietal, lateral occipital and superior parietal cortices, with an effect size of Cohen's $d = -0.14$. Lower cortical thickness was also observed in the orbitofrontal cortex (OFC), anterior and posterior cingulate cortices, insula, precentral and postcentral gyri, and several temporal and occipital areas, with effect sizes ranging from Cohen's $d = -0.06$ to -0.13. These differences were predominantly bilateral, consistent with previous findings that showed no significant asymmetries in cortical thickness,

https://doi.org/10.1038/s44220-026-00667-9

surface area or subcortical volumes between patients with MDD and HCs[13].

Our current findings align with previous meta-analytic studies. In a prior ENIGMA MDD study, which used FreeSurfer regions of interest from the Desikan–Killiany atlas rather than vertex-based FreeSurfer measures, we similarly identified lower cortical thickness in regions including the anterior and posterior cingulate cortices, insula, frontal gyri, OFC, fusiform gyrus, and middle and inferior temporal gyrus, with effect sizes ranging from Cohen's d = −0.10 to −0.14 (ref. 7). Other meta-analyses have also reported lower cortical thickness in MDD in areas including the inferior frontal gyrus and parietal cortex[5,14,15]. The posterior cingulate cortex, anterior cingulate cortex (ACC), inferior parietal cortex, inferior frontal gyrus and precuneus contribute to the default mode network, in which alterations have been consistently found in patients with depression from both mixed/white-majority[16] and Asian[17] populations and have been linked to rumination and dysfunctional self-referential processes characteristic of depression[18]. Notably, the subgenual ACC, part of the rostral ACC, plays a key role in emotion regulation and has emerged as a core hub for guiding functional connectivity in personalized transcranial magnetic stimulation protocols targeting the dorsolateral prefrontal cortex, particularly for treatment-resistant depression[19]. Many of the regions implicated in our study connect to a larger corticolimbic system (implicating the prefrontal cortex, temporal cortex and limbic areas), which provides a substrate for adaptive behavioral and emotional responses. Alterations observed in the occipital cortex, including the lingual gyrus, may point to a selective perceptual bias in processing of emotional information[20]. Notably, lower thickness of the OFC has been one of the most consistent findings across large-scale meta-analyses[7,14]. The OFC is implicated in emotion processing and reward, with dysfunction in the lateral OFC linked to bias toward negative emotional stimuli and dysfunction in the medial OFC linked to anhedonia and deficits in reward anticipation[21].

We also identified areas of higher cortical thickness in individuals with MDD, specifically in regions including the pericalcarine and lingual gyri bilaterally, and the left cuneus, with effect sizes ranging from Cohen's d = 0.08 to 0.09. The pericalcarine region is the initial region of visual processing, and the lingual gyrus is associated with high-level visual processing and visual memory. These findings might reflect heightened processing demands of aversive stimuli and fear or emotional learning in these areas, and aberrant visual–emotional integration in MDD[22,23].

In terms of underlying neurobiological mechanisms associated with the observed cortical thickness alterations, a recent 'virtual histology approach' linked alterations in cortical thickness to variations in expressions of genes related to pyramidal CA1 cells, implicated in the remodeling of dendrites that play a critical role in neural communication[24]. Remodeling of dendrites can result from environmental factors (for example, stress) and genetic influences. Since cortical thickness seems to be less influenced by genetic factors than surface area, which is associated with three times more genome-wide significant loci[25], the observed alterations in cortical thickness may be driven by remodeling of dendrites as a result of negative environmental factors (for example, the effects of having long-term stress caused by the disorder).

Within adults with MDD, those with acute depression appeared to drive the observed cortical thickness alterations, with larger effect sizes up to Cohen's d = −0.19. Adults in remission did not show significant differences compared to controls; however, they also did not differ significantly from acutely depressed patients, suggesting that remitted patients have intermediate cortical thickness between acutely depressed patients and HCs. The importance of acute symptoms was further highlighted by an association between greater depressive symptom severity (measured by the HDRS) and lower cortical thickness in the left caudal middle frontal, precentral, and superior temporal gyri (Cohen's d = −0.08 to −0.09) in adults with MDD. Similar patterns and effect sizes of cortical thickness reductions were seen in adult patients with a first episode of MDD and with recurrent episodes compared to HCs, suggesting that the stage of illness did not significantly influence the extent of cortical thickness reductions in MDD. By contrast, widespread reductions in cortical thickness were observed in adults who were using antidepressant medication at the time of the scan, compared to both adult patients not using antidepressants and HCs (Cohen's d = −0.08 to −0.17). Although antidepressants are generally believed to have neuroprotective effects, such as promoting brain-derived neurotrophic factor[26], this pattern may reflect confounding by clinical characteristics. In our data, current symptom severity at the time of scanning, indexed by HDRS, is associated with lower cortical thickness. However, harmonized measures of illness chronicity, such as episode duration and number of episodes, were not consistently available across cohorts. Therefore, the cross-sectional design and limited clinical characterization preclude definitive conclusions regarding medication effects[27].

The alterations in cortical thickness were significant in adults with MDD, whereas no differences in cortical thickness were observed in adolescents with MDD. However, the lack of significant cortical thickness findings in adolescents may be due to lower statistical power in the adolescent group (n = 908 adolescents with MDD and n = 883 adolescent controls) compared to the adult group (n = 4,139 adults with MDD and n = 5,190 adult controls). Notably, the spatial patterns of thickness abnormalities and effect sizes were similar, or even greater, in adolescents compared to adults (Supplementary Fig. 11). In addition, the effect size for case–control differences weakened with age in the region with the highest effect size, specifically the cluster spanning the left superior parietal, lateral occipital, and inferior parietal areas, which also suggests that cortical thickness alterations may be more pronounced in younger individuals with MDD. Larger populations of adolescents with MDD are required to confirm this. The finding that MDD effects weakened with age possibly suggests that normative age-related thinning may progressively reduce the 'gap' between individuals with MDD and HCs. With increasing age, other processes that were not accounted for in this study and can affect cortical thickness, such as cerebral small vessel disease[28], may come into play in HCs.

While rates of depression are higher in female than male individuals[29], no sex-by-diagnosis interactions were observed in this study. Although some studies have reported sex differences in brain structure among individuals with MDD, most of this evidence comes from small populations[30] and has not been replicated in larger studies[5,7]. This suggests that cortical thickness alterations associated with MDD may be comparable across male and female individuals and that structural brain differences are unlikely to account for the higher rates of depression observed in female individuals.

Unlike our present study, which found no differences in surface area in adults and adolescents with MDD, our previous ENIGMA MDD study observed decreased surface area in frontal, parietal and occipital regions in adolescents with MDD, with effect sizes up to Cohen's d = −0.42 (ref. 7). In addition, the COCORO consortium reported lower surface area in adult patients with MDD (aged >21 years)[5]. The larger cohort size in this study probably provides more stable effect sizes and accurate estimates of surface area differences because higher-powered studies have a greater probability of finding 'true' effects and produce less false negatives, and larger cohorts are often associated with increased population heterogeneity, which can inflate overall variance contributing to lower effect sizes[31–33]. Nonetheless, further research should evaluate whether potential subgroups of people with MDD exist that show surface area alterations. This may represent an early developing subtype of depressive disorder, shaped by genetic factors or early life adversity[34,35], given that, compared to cortical thickness, cortical surface area has a higher genetic heritability and a genetic correlation with MDD and depressive symptoms (this genetic association is absent for cortical thickness)[25]. Cortical surface area is also

 https://doi.org/10.1038/s44220-026-00667-9

determined earlier in development, and is less strongly affected by later environmental influences[36,37].

Beyond limited statistical power for observing differences in cortical surface area and thickness in adolescents with MDD, lack of effects in adolescents may also be related to developmental effects. From early childhood, cortical thickness undergoes substantial thinning that continues into adulthood, with regionally variable rates of change[27,38]. Surface area undergoes rapid expansion in infancy, followed by slower growth throughout childhood, with peaks estimated around age 11 years, after which it slowly decreases[39]. Our population of younger participants spanned from childhood to late adolescence, and therefore it is possible that normative developmental variability in cortical maturation may have masked disorder-related alterations. Future studies with larger cohorts of adolescents that examine distinct age groups separately, as well as longitudinal studies, are needed to understand the contribution of developmental effects and depression-related changes in cortical morphology.

When analyzed separately, DIRECT consortium participants with MDD showed more widespread cortical thinning than ENIGMA MDD consortium participants. The proportion of white participants also moderated the effect size of the case–control differences, with a higher proportion of white participants associated with greater cortical thickness in paracentral, precentral and inferior parietal regions. Importantly, these biological differences, rather than solely being attributed to 'race' or 'ethnicity' per se, may reflect underlying social and political factors[40]. For example, the more widespread effects of MDD-related differences in cortical thickness among individuals with an Asian background may reflect variations in help-seeking patterns, as those seeking psychiatric services may be a group with more severe depressive psychopathology[41].

The effect sizes of the observed cortical thickness alterations were small overall (Cohen's $d < 0.20$), consistent with previous large-scale studies using harmonized clinical and imaging protocols[6,32]. Thus, variability in individual measures of structural brain alterations accounts for only a small percentage of the explained variance in the complex depression phenotype[32]. This is in line with the genetics literature, where polygenic risk scores explain up to 3% of variance[42]. Future studies should examine depression subtypes with more pronounced structural brain alterations and consider shifting from a focus on structural brain measures as diagnostic biomarkers to prognostic or predictive markers of, for instance, relapse/recurrence or treatment response.

Strengths of this study include its large sample size, providing robust statistical power, which enhances the reliability and precision of our findings[43]. This increased power enables the detection of smaller effect sizes that may not be evident in individual studies with limited sample sizes. Moreover, we used harmonized processing and statistical analysis protocols across all 64 cohorts included in this study, thereby reducing statistical heterogeneity and researcher degrees of freedom, and ensuring reproducibility. In addition, combining data from the ENIGMA MDD and DIRECT consortia, which include studies from very diverse geographic locations with diverse cultures, ensures that our results are generalizable across different populations, mitigating the risk of location-specific biases. By integrating data from various regions and cultures, our study provides more nuanced and globally relevant findings. We refrained from examining the potential impact of race, ethnicity or geographic location on the case–control differences, because it is impossible to distinguish whether such effects represent biological differences, differences in social and political factors, or even variations in help-seeking patterns and access to mental health care.

Nonetheless, our study also has limitations, including heterogeneity in phenotyping across individual studies including clinical assessments, limiting the analysis of sources of clinical heterogeneity. Our meta-regression analyses also showed that FreeSurfer version and field strength impacted the effect sizes for case–control differences in cortical thickness, although only in a few regions. As some participating sites were restricted in terms of data sharing, we used a meta-analytic framework rather than a mega-analysis. Prior work indicates that when preprocessing (and, for meta-analyses, the within-site group comparisons) are harmonized across sites, disease-related effect sizes based on FreeSurfer measures are highly similar for mega- and meta-analytic approaches; however, mega-analyses can yield smaller standard errors and thus narrower confidence intervals[44]. Furthermore, as our findings are based on cross-sectional studies, they require further investigation in longitudinal studies to elucidate, for example, influences of brain development and aging, medication effects and the clinical relevance of the observed structural brain alterations in MDD.

Conclusions

This extensive individual participant-based meta-analysis of data from 64 cohorts from the ENIGMA MDD and DIRECT consortia reveals widespread cortical thickness reductions in individuals with MDD, notably in prefrontal, cingulate, temporal, parietal and occipital areas, along with increased thickness in pericalcarine and lingual gyri and the cuneus. These differences are primarily driven by acute depressive episodes, an adult age of onset and antidepressant medication use in adults. Despite small effect sizes (mostly $|\text{Cohen's } d| < 0.20$) and heterogeneity in data acquisition across cohorts, the large and diverse sample size enhances the reliability and global relevance of these findings. Future research should focus on longitudinal studies to understand the evolution of these changes, explore potential depression subtypes and investigate prognostic markers for clinical outcomes and treatment responses.

Methods

Cohorts

The DIRECT Consortium comprises 23 Chinese groups that have provided neuroimaging and clinical data from both patients with MDD and HCs[12]. The ENIGMA MDD consortium includes 41 international study cohorts with neuroimaging and clinical data from patients with MDD and HCs[8]. A table of the participating sites may be found in Supplementary Tables 1 and 2. Our study, which involved a total of 12,274 participants, analyzed data from 5,736 patients with MDD and 6,538 HCs. Demographic information for each cohort can be found in Supplementary Table 2, while clinical characteristics are provided in Supplementary Table 3. All participating sites received approval from their respective institutional review boards and ethics committees, and all study participants provided written informed consent. The specific instruments used for diagnosis are presented in Supplementary Table 4.

Image processing and analysis

Three-dimensional volumetric structural T1-weighted (T1w) magnetic resonance imaging brain scans were acquired and analyzed locally at each site using harmonized analysis and quality control protocols provided by the DIRECT and ENIGMA MDD consortia, respectively. For the DIRECT consortium, FreeSurfer[45] (version 6.0.1) with the DPABISurf[46] (version 1.5) interface (the fMRIPrep[47] (version 20.2.1) docker) was used to perform all cortical constructions. Specifically, the T1w image was corrected for intensity nonuniformity with N4BiasFieldCorrection[48], distributed with ANTs 2.2.0[49] (RRID: SCR_004757). The T1w reference was then skull-stripped with a Nipype implementation of the antsBrainExtraction.sh workflow (from ANTs), using OASIS30ANTs as a target template. Brain tissue segmentation of cerebrospinal fluid, white matter and GM was performed on the brain-extracted T1w using fast (FSL 5.0.9, RRID: SCR_002823)[50]. Brain surfaces were reconstructed using recon-all (FreeSurfer 6.0.1, RRID: SCR_001847)[45], and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical GM of Mindboggle (RRID: SCR_002438)[51]. For the ENIGMA MDD consortium, FreeSurfer (versions 5–7, with the version effect analyzed later) was used to perform all cortical constructions.

The image acquisition parameters and software descriptions can be found in Supplementary Table 5. Vertex-wise cortical thickness and surface area maps were registered to fsaverage space with FreeSurfer and further smoothed with a 10-mm full width at half maximum kernel before entering statistical analyses.

Cohort selection

From 12,274 participants (5,736 patients with MDD and 6,538 HCs), we selected participants for group statistical analyses through the following criteria: (1) Sites Cardiff, MODECT and SanRaffaele were excluded in patient versus control analyses since they only contained patients with MDD, resulting in 5,475 patients with MDD and 6,538 HCs, (2) then we removed the CSAN site with fewer than 10 participants in either group (10 was selected arbitrarily to balance the objectives of optimizing overall sample size and minimizing extreme biases), resulting in 5,417 patients with MDD and 6,535 HCs, (3) participants with bad imaging data and bad surface reconstruction (by visual inspection) were excluded, resulting in 5,404 patients with MDD and 6,527 HCs, (4) next, for DIRECT sites, we calculated the average value of the thickness maps for all participants and correlated each individual's cortical thickness map with the average map. The threshold was set using the mean of all correlation coefficients minus two times the standard deviation, resulting in $r = 0.47$. We removed 145 participants with coefficients below this threshold, 13 of whom were previously excluded. Most participants in IS008 and IS020 also had coefficients below $r = 0.47$, leading to the removal of these two sites, leaving 5,316 patients with MDD and 6,454 HCs. (5) In addition, we excluded those sites that did not fit in the model, that is, we excluded sites DCHS, MoralDilemma and StanfordFAA, as these sites only contain female individuals and thus did not fit in the general model with sex as covariates. After this exclusion, 5,256 patients with MDD and 6,320 HCs passed the quality control and cohort selection procedure.

Statistical framework for meta-analysis

Multiple linear regression models were used to analyze group differences in vertex-wise cortical thickness and surface area between patients and controls within each local cohort. The predictor of interest was a binary indicator of diagnosis (patients with MDD, 1; controls, −1) and all models were adjusted for age and sex. In the statistical analysis for surface area, the estimated total intracranial volume was also used as a covariate. For models testing sex-by-diagnosis and age-by-diagnosis interactions, a multiplicative predictor was the predictor of interest, and the main effect of each predictor was included in the model using the same procedure. Within-site t values were converted into Cohen's d and entered into a random-effect meta-model (using R metansue; https://www.metansue.com/) for voxel-wise meta-analysis (integrated as y_Meta_Image_CallR.m in DPABI).

As we found differences in patterns of cortical thickness and surface area brain alterations between adolescent and adult patients with MDD in our prior work[7], we conducted separate meta-analyses for adolescent participants (age ≤21 years) and adult participants (age >21 years)[7]. For the adult analysis, we further stratified patients on the basis of their age at illness onset (adolescent onset, ≤21 years; adult onset, >21 years[52]) to investigate the effects of illness onset. Both in adolescents and adult participants, we also split patients on the basis of the stage of their illness (first episode versus recurrent), as well as antidepressant use at the time of their scan (yes/no) and remission status (acutely depressed versus in remission). The associations between symptom severity at the time of scanning, as measured by the 17-item HDRS, and cortical thickness and surface area were also examined. Information on the included cohorts and the total sample sizes for each model can be found in the Main text. Throughout the Analysis, we provide P values corrected for multiple comparisons using the Benjamini–Hochberg procedure[53] (separate for cortical thickness and cortical surface area), ensuring a vertex-wise FDR limited to 5% across all cortical vertices in the fsaverage space (left and right hemispheres were combined to $n = 299,881$ vertices), to control for multiple comparisons across the entire cortical surface.

Moderator analyses with meta-regression

In addition, we conducted moderator analyses using meta-regression analyses to determine whether individual site characteristics accounted for a significant portion of the variance in effect sizes across sites in the meta-analyses. We tested whether the variance in effect sizes across sites in the meta-analyses could be accounted for by the following individual site characteristics: FreeSurfer version (version 6 versus other versions (mostly version 5 with one site exception of version 7), scanner vendor (other scanner versus Siemens, and GE versus Siemens), field strength (1.5 T versus 3 T) and type of cohort (general population versus clinical). We additionally included the proportion of white, Asian and African participants at each site as moderators to examine whether site-level ethnic composition contributed to between-site variability.

For the DIRECT datasets, approval was obtained from the Ethics Committee of the Institute of Psychology, Chinese Academy of Sciences. For ENIGMA MDD, ethics approval for the storage and management of consortium data was granted by the University of Melbourne Human Research Ethics Office. All participating sites obtained approval from their respective local institutional review boards or ethics committees, and all participants provided written informed consent.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The dataset required to interpret, replicate and extend the findings of this study consists of the derived meta-analytic statistical maps and summary data generated during the analyses for all participating sites. These data are available at https://doi.org/10.57760/sciencedb.29686. Individual-level participant data from the contributing cohorts cannot be publicly shared owing to local ethics and data-sharing restrictions across participating sites.

Code availability

Custom scripts used for data processing, statistical analysis and figure generation in this study are publicly available via GitHub at https://github.com/Chaogan-Yan/PaperScripts/tree/master/Yan_2026_NatureMentalHealth.

References

1. Smith, K. Mental health: a world of depression. *Nature* **515**, 180–181 (2014).
2. Cuijpers, P. et al. Absolute and relative outcomes of psychotherapies for eight mental disorders: a systematic review and meta-analysis. *World Psychiatry* **23**, 267–275 (2024).
3. Trivedi, M. H. et al. Evaluation of outcomes with citalopram for depression using measurement-based care in STAR*D: implications for clinical practice. *Am. J. Psychiatry* **163**, 28–40 (2006).
4. Wang, K. et al. Brain structural abnormalities in adult major depressive disorder revealed by voxel- and source-based morphometry: evidence from the REST-meta-MDD Consortium. *Psychol. Med.* **53**, 3672–3682 (2023).
5. Matsumoto, J. et al. Cerebral cortical structural alteration patterns across four major psychiatric disorders in 5549 individuals. *Mol. Psychiatry* **28**, 4915–4923 (2023).
6. Kochunov, P. et al. Comparison of regional brain deficit patterns in common psychiatric and neurological disorders as revealed by big data. *NeuroImage Clin.* **29**, 102574 (2021).
7. Schmaal, L. et al. Cortical abnormalities in adults and adolescents with major depression based on brain scans from 20 cohorts

worldwide in the ENIGMA Major Depressive Disorder Working Group. *Mol. Psychiatry* **22**, 900–909 (2017).
8. Schmaal, L. et al. ENIGMA MDD: seven years of global neuroimaging studies of major depression through worldwide data sharing. *Transl. Psychiatry* **10**, 172 (2020).
9. Schmaal, L. et al. Subcortical brain alterations in major depressive disorder: findings from the ENIGMA Major Depressive Disorder Working Group. *Mol. Psychiatry* **21**, 806–812 (2016).
10. Desikan, R. S. et al. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *NeuroImage* **31**, 968–980 (2006).
11. Ho, T. C. et al. Subcortical shape alterations in major depressive disorder: findings from the ENIGMA Major Depressive Disorder Working Group. *Hum. Brain Mapp.* **43**, 341–351 (2022).
12. Chen, X. et al. Subgenual anterior cingulate cortex functional connectivity abnormalities in depression: insights from brain imaging big data and precision-guided personalized intervention via transcranial magnetic stimulation. *Sci. Bull.* https://doi.org/10.1016/j.scib.2025.05.042 (2025).
13. de Kovel, C. G. F. et al. No alterations of brain structural asymmetry in major depressive disorder: an ENIGMA Consortium Analysis. *Am. J. Psychiatry* **176**, 1039–1049 (2019).
14. Suh, J. S. et al. Cortical thickness in major depressive disorder: a systematic review and meta-analysis. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **88**, 287–302 (2019).
15. Li, Q. et al. Meta-analysis of cortical thickness abnormalities in medication-free patients with major depressive disorder. *Neuropsychopharmacology* **45**, 703–712 (2020).
16. Kaiser, R. H., Andrews-Hanna, J. R., Wager, T. D. & Pizzagalli, D. A. Large-scale network dysfunction in major depressive disorder: a meta-analysis of resting-state functional connectivity. *JAMA Psychiatry* **72**, 603–611 (2015).
17. Yan, C. G. et al. Reduced default mode network functional connectivity in patients with recurrent major depressive disorder. *Proc. Natl Acad. Sci. USA* **116**, 9078–9083 (2019).
18. Zhou, H. X. et al. Rumination and the default mode network: meta-analysis of brain imaging studies and implications for depression. *NeuroImage* **206**, 116287 (2020).
19. Fox, M. D., Buckner, R. L., White, M. P., Greicius, M. D. & Pascual-Leone, A. Efficacy of transcranial magnetic stimulation targets for depression is related to intrinsic functional connectivity with the subgenual cingulate. *Biol. Psychiatry* **72**, 595–603 (2012).
20. Kanwisher, N. & Yovel, G. The fusiform face area: a cortical region specialized for the perception of faces. *Phil. Trans. R. Soc. B* **361**, 2109–2128 (2006).
21. Zhang, B. et al. Roles of the medial and lateral orbitofrontal cortex in major depression and its treatment. *Mol. Psychiatry* **29**, 914–928 (2024).
22. Du, M. et al. Brain grey matter volume alterations in late-life depression. *J. Psychiatry Neurosci.* **39**, 397–406 (2014).
23. Ancelin, M. L. et al. Lifetime major depression and grey-matter volume. *J. Psychiatry Neurosci.* **44**, 45–53 (2019).
24. Patel, Y. et al. Virtual histology of cortical thickness and shared neurobiology in 6 psychiatric disorders. *JAMA Psychiatry* **78**, 47–63 (2021).
25. Grasby, K. L. et al. The genetic architecture of the human cerebral cortex. *Science* https://doi.org/10.1126/science.aay6690 (2020).
26. Castrén, E. & Monteggia, L. M. Brain-derived neurotrophic factor signaling in depression and antidepressant action. *Biol. Psychiatry* **90**, 128–136 (2021).
27. Frangou, S. et al. Cortical thickness across the lifespan: data from 17,075 healthy individuals aged 3–90 years. *Hum. Brain Mapp.* **43**, 431–451 (2022).
28. Petersen, M. et al. Mapping the association of cerebral small vessel disease, gray matter integrity, and cognitive function. *Neurology* **105**, e214215 (2025).
29. Albert, P. R. Why is depression more prevalent in women? *J. Psychiatry Neurosci.* **40**, 219–221 (2015).
30. Mohammadi, S. et al. Brain-based sex differences in depression: a systematic review of neuroimaging studies. *Brain Imaging Behav.* **17**, 541–569 (2023).
31. Miller, K. L. et al. Multimodal population brain imaging in the UK Biobank prospective epidemiological study. *Nat. Neurosci.* **19**, 1523–1536 (2016).
32. Winter, N. R. et al. Quantifying deviations of brain structure and function in major depressive disorder across neuroimaging modalities. *JAMA Psychiatry* **79**, 879–888 (2022).
33. Marek, S. et al. Reproducible brain-wide association studies require thousands of individuals. *Nature* **603**, 654–660 (2022).
34. Jha, S. C. et al. Environmental influences on infant cortical thickness and surface area. *Cereb. Cortex* **29**, 1139–1149 (2019).
35. Tozzi, L. et al. Interactive impact of childhood maltreatment, depression, and age on cortical brain structure: mega-analytic findings from a large multi-site cohort. *Psychol. Med.* **50**, 1020–1031 (2020).
36. Fjell, A. M. et al. Continuity and discontinuity in human cortical development and change from embryonic stages to old age. *Cereb. Cortex* **29**, 3879–3890 (2019).
37. Storsve, A. B. et al. Differential longitudinal changes in cortical thickness, surface area and volume across the adult life span: regions of accelerating and decelerating change. *J. Neurosci.* **34**, 8488–8498 (2014).
38. Ducharme, S. et al. Trajectories of cortical thickness maturation in normal brain development—the importance of quality control procedures. *NeuroImage* **125**, 267–279 (2016).
39. Bethlehem, R. A. I. et al. Brain charts for the human lifespan. *Nature* **604**, 525–533 (2022).
40. Fu, J. et al. Cross-ancestry genome-wide association studies of brain imaging phenotypes. *Nat. Genet.* **56**, 1110–1120 (2024).
41. Bigdeli, T. B. et al. Genetic effects influencing risk for major depressive disorder in China and Europe. *Transl. Psychiatry* **7**, e1074 (2017).
42. Howard, D. M. et al. Genome-wide meta-analysis of depression identifies 102 independent variants and highlights the importance of the prefrontal brain regions. *Nat. Neurosci.* **22**, 343–352 (2019).
43. Algermissen, J. & Mehler, D. M. A. May the power be with you: are there highly powered studies in neuroscience, and how can we get more of them. *J. Neurophysiol.* **119**, 2114–2117 (2018).
44. Boedhoe, P. S. W. et al. An empirical comparison of meta- and mega-analysis with data from the ENIGMA Obsessive–Compulsive Disorder Working Group. *Front. Neuroinform.* **12**, 102 (2018).
45. Dale, A. M., Fischl, B. & Sereno, M. I. Cortical surface-based analysis. I. Segmentation and surface reconstruction. *NeuroImage* **9**, 179–194 (1999).
46. Yan, C.-G., Wang, X.-D. & Lu, B. DPABISurf: data processing and analysis for brain imaging on surface. *Sci. Bull.* **66**, 2453–2455 (2021).
47. Esteban, O. et al. fMRIPrep: a robust preprocessing pipeline for functional MRI. *Nat. Methods* **16**, 111–116 (2019).
48. Tustison, N. J. et al. N4ITK: improved N3 bias correction. *IEEE Trans. Med. Imaging* **29**, 1310–1320 (2010).
49. Avants, B. B., Epstein, C. L., Grossman, M. & Gee, J. C. Symmetric diffeomorphic image registration with cross-correlation:

evaluating automated labeling of elderly and neurodegenerative brain. *Med. Image Anal.* **12**, 26–41 (2008).
50. Zhang, Y., Brady, M. & Smith, S. Segmentation of brain MR images through a hidden Markov random field model and the expectation-maximization algorithm. *IEEE Trans. Med. Imaging* **20**, 45–57 (2001).
51. Klein, A. et al. Mindboggling morphometry of human brains. *PLoS Comput. Biol.* **13**, e1005350 (2017).
52. Benazzi, F. Classifying mood disorders by age-at-onset instead of polarity. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **33**, 86–93 (2009).
53. Benjamini, Y. & Hochberg, Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. B* **57**, 289–300 (1995).

Acknowledgments

We acknowledge the following: Brain Science and Brain-Like Intelligence Technology-National Science and Technology Major Project (grant nos. 2021ZD0200600 and 2021ZD0200601 to C.-G.Y.), National Natural Science Foundation of China (grant nos. 82572192 and 82122035 to C.-G.Y.), Beijing Nova Program of Science and Technology (grant no. 20230484465 to C.-G.Y.), Beijing Natural Science Foundation (grant no. J230040 to C.-G.Y.), National Health and Medical Research Council (NHMRC) (App 1161356 to B.C.-D.), German Research Foundation (grant no. STR 1146/18-1 to B.S.M.), DFG SFB/TRR 393 consortium (grant no. 521379614 to B.S.), LOEWE program of the Hessian Ministry of Science and Arts (grant no. LO EWE1/16/519/03/09.001(0009)/98 to B.S.), Intramural Research Program at the National Institute of Mental Health, National Institutes of Health (grant no. ZIAMH002857 to C.A.Z.), National Institute for Health Research Biomedical Research Centre for Mental Health at South London and Maudsley NHS Foundation Trust and King's College London (C.H.Y.F.), National Institute of Mental Health (grant no. R01MH134236 to C.H.Y.F.), Milken Institute Baszucki Brain Research (grant no. BD00029 to C.H.Y.F.), Rosetrees Trust (grant no. CF20212104 to C.H.Y.F.), International Psychoanalytical Association (grant no. IPA158102845 to C.H.Y.F.), National Institute for Health Research Biomedical Research Centre for Mental Health at South London and Maudsley NHS Foundation Trust and King's College London (D.D.), Research Wales (grant no. HS/14/20 to D.E.J.L.), Medical Research Council (grant no. G 1100629 to D.E.J.L.), Agence Nationale de la Recherche (ANR) (grant ANR SAMENTA 2012 to E.C.), Institut National de la Santé Et de la Recherche Médicale (grant no. C1325 to E.C.), German Research Foundation SFB/TRR 393 (grant no. 521379614 to E.J.L.), Federal Ministry of Education and Research (grant nos. 01ZZ9603, 01ZZ0103 and 01ZZ0403 to H.J.G.), Ministry of Cultural Affairs (H.J.G.), Social Ministry of the Federal State of Mecklenburg-West Pomerania (H.J.G.), NHMRC Investigator Grant (grant no. 2016346 to I.B.H.), National Institute of Mental Health (grant no. R37MH101495 to I.H.G.), National Institute of Mental Health (grant no. 1R01MH085667-01A1 to J.C.S.), John S. Dunn Foundation (J.C.S.), Pat Rutherford Chair in Psychiatry (J.C.S.), Intramural Research Program at the National Institute of Mental Health, National Institutes of Health (grant no. ZIAMH002857 to J.W.E.), German Federal Ministry for Education and Research (BMBF) (grant nos. FKZ-01ER0816 and FKZ-01ER1506 to K.B.), National Institute of Mental Health (grant no. K23MH090421 to K.R.C.), National Alliance for Research on Schizophrenia and Depression (K.R.C.), University of Minnesota Graduate School (K.R.C.), Minnesota Medical Foundation (K.R.C.), Biotechnology Research Center (grant no. P41 RR008079 to K.R.C.), Deborah E. Powell Center for Women's Health Seed Grant (K.R.C.), National Institute of Mental Health (grant no. R01MH125850 to M.D.S.), Federal Ministry of Education and Research (grant nos. 01ZZ9603, 01ZZ0103 and 01ZZ0403 to R.B.), Ministry of Cultural Affairs (R.B.), Social Ministry of the Federal State of Mecklenburg-West Pomerania (R.B.), Siemens Healthineers (R.B.), Federal State of Mecklenburg-West Pomerania (R.B.), Agence Nationale de la Recherche (ANR) (grant ANR SAMENTA 2012 to R.C.), Institut National de la Santé Et de la Recherche Médicale (grant no. C1325 to R.C.), German Federal Ministry of Education and Research (BMBF) (grant no. 01 ZX 1507 to R.G.-M.), NIH (grant nos. R01MH123163, R01MH121246 and R01MH116147 to S.I.T.), NIH Big Data to Knowledge program (grant no. U54 EB020403 to P.M.T.), NHMRC (grant no. APP2025674 to S. Medland), NHMRC of Australia (grant no. 1125504 to S.W.), Brain and Behavior Research Foundation (S.W.), National Institutes of Health (grant no. K01MH117442 to T.C.H.), Klingenstein Third Generation Foundation (T.C.H.), Stanford Maternal Child and Health Institute (T.C.H.), Stanford Center for Cognitive and Neurobiological Imaging Center (T.C.H.), Ray and Dagmar Dolby Family Fund (T.C.H.), National Institutes of Health (grant nos. R01MH127176 and R21MH130817 to T.C.H.), German Federal Ministry of Education and Research (BMBF) (grant no. 01 ZX 1507 to T.E.-G.), German Research Foundation (DFG) (grant nos. FOR2107 KI588/14-1, KI588/14-2, KI588/20-1 and KI588/22-1 to T. Kircher), DFG SFB/TRR 393 consortium (grant no. 521379614 to T. Kircher), LOEWE program of the Hessian Ministry of Science and Arts (grant no. LOE WE1/16/519/03/09.001(0009)/98 to T. Kircher), National Center for Complementary and Integrative Health (grant nos. 1R61AT009864-01A1 and R33AT009864 to T.T.Y.), German Research Foundation (DFG) (grant nos. FOR2107 DA1151/5-1, DA1151/5-2, DA1151/9-1, DA1151/10-1 and DA1151/11-1 to U.D.), DFG SFB/TRR 393 consortium (project grant no. 521379614 to U.D.), Interdisciplinary Center for Clinical Research (IZKF) of the Medical Faculty of Münster (grant no. Dan3/016/26 to U.D.) Convergence|Healthy Start (Y.T.), National Institute of Mental Health (grant no. K23MH090421 to Z.B.), National Alliance for Research on Schizophrenia and Depression (Z.B.), University of Minnesota Graduate School (Z.B.), Minnesota Medical Foundation (Z.B.), Biotechnology Research Center (grant no. P41 RR008079 to Z.B.) and Deborah E. Powell Center for Women's Health Seed Grant (Z.B.), National Natural Science Foundation of China (grant nos. 82572350 to T.-L.C.). We would like to acknowledge the late Professor Dan J. Stein for his longstanding leadership and collaboration within ENIGMA-MDD Working Group and for his invaluable contributions to psychiatry, neuroscience, and mental health research. His intellectual generosity, mentorship, and unwavering support shaped this work and the broader field in lasting ways. His scientific legacy and his kindness to colleagues continue to inspire us.

Author contributions

C.-G.Y., Z.-H.W., E.P. and L.S. conceived and designed the study and interpreted the data. C.-G.Y., L.K.M.H., N.A., D.A., Z.B., S.E.E.C.B., J.B., F.B., K.B., B.B., T.B., R.B., L.-P.C., G.-M.C., J.-S.C., S.-C.C., T.C., T.-L.C., X.C., Y.-R.C., Y.-Q.C., Z.-S.C., R.C., C.G.C., E.C., B.C.-D., X.-L.C., K.R.C., U.D., C.D., D.D., A.D., T.E.-G., J.W.E., K.F., C.H.Y.F., P.F.-C., Q.-L.G., Q.-Y.G., A.S.G., I.H.G., R.G.-M., H.J.G., N.G., D.G., W.-B.G., T.H., G.B.H., J.P.H., B.H., S.H., C.-C.H., M.H., I.B.H., T.C.H., Z.-J.-Y.H., Q.H., N.J., H.J., A.J., X.-L.J., F.-N.J., T. Kamishikiryo, T. Kircher, S.-M.K., A. Kraus, A. Krug, L.K., J.L., T. Lancaster, N.I.L., E.J.L., M.L., B.-J.L., F.L., H.-X.L., T. Li, X.-Y. Li, Y.-F.L., Z.-D.L., D.E.J.L., H.-L.L., X.-Y. Liu, Z.-N.L., Y.-S.L., Y.-C.L., J.-P.L., B.L., F.P.M., L.A.M., A.M.M., S. Medland, D.M.A.M., S. Meinert, B.M., A.O., G.O., M.L.O., B.W.J.H.P., S.P., E.P.-C., M.J.P., J.Q., L.R., H.S.R., E.R.-C., M.D.S., R.S., H. Schlieper, A.S., X.-X.S., H. Shinzato, T.-M.S., K.S., J.C.S., D.J.S., B.S., L.T.S., P.-F.S., F.T.-O., S.I.T., P.M.T., Y.T., A.U., N.J.A.W., S.J.A.V.W., I.M.V., D.J.V., H.V., M. Wagenmakers, H.W., M. Walter, C.-Y.W., H.-L.W., H.-N.W., X.W., X.-Y.W., Y.W., Y.-W.W., L.T.W., H.C.W., S.W., N.R.W., K.W., M.-J.W., X.-P.W., X.-R.W., Y.-K.W., C.-N.W., C.-M.X., G.-R.X., P.X., X.-F.X., Z.-P.X., T.T.Y., H. Yang, J.Y., H. Yu, Y.-Q.Y., M.-L.Y., Y.-G.Y., Y.-F.Z., C.A.Z., A.-X.Z., K.-R.Z., W.Z., J.-P.Z., W.-Y.Z., J.-J.Z., G.B.Z.S., X.-N.Z., E.P. and L.S. contributed to data collection, data processing and/or data provision from the participating cohorts. All authors critically revised the paper and approved the final version for publication.

 https://doi.org/10.1038/s44220-026-00667-9

Competing interests

For the DIRECT consortium, the authors declare no competing interests. Regarding ENIGMA MDD, H.J.G. has received travel grants and speakers honoraria from Neuraxpharm, Servier, Indorsia and Janssen Cilag. I.B.H. has previously led community-based and pharmaceutical industry-supported (Wyeth, Eli Lily, Servier, Pfizer and AstraZeneca) projects focused on the identification and better management of anxiety and depression. He is the chief scientific advisor to, and a 3.2% equity shareholder in, InnoWell Pty Ltd. InnoWell was formed by the University of Sydney (45% equity) and PwC (Australia; 45% equity) to deliver the $30 million Australian government-funded Project Synergy (2017–2020) and to lead transformation of mental health services internationally through the use of innovative technologies. J.C.S. has received funding from other sources, and potential conflict of interests include: Alkermes (advisory board), Boehringer Ingelheim (consultant), Compass Pathways (research grant), Johnson and Johnson (consultant), LivaNova (consultant), Relmada (research grant), Sunovion (research grant) and Mind Med (research grant). T.T.Y.: UCSF and NeuroQore have a clinical trials agreement that provides funds to UCSF, and a portion of these funds are provided to T.T.Y. for research collaboration and papers with NeuroQore. None of the funds from NeuroQore were used to support T.T.Y. for this paper. T.T.Y. is a professor and full-time UCSF faculty member, and he is paid by UCSF. T. Kircher received unrestricted educational grants from Servier, Janssen, Recordati, Aristo, Otsuka and Neuraxpharm.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s44220-026-00667-9.

Correspondence and requests for materials should be addressed to Chao-Gan Yan or Lianne Schmaal.

Peer review information *Nature Mental Health* thanks Xujun Duan, Delin Sun and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Chao-Gan Yan[1,2,3] ✉, Zi-Han Wang[2,4], Laura K. M. Han[5,6,7], Nina Alexander[8], Dag Alnæs[9,10], Zeynep Başgöze[11,12], Stéphanie E. E. C. Bauduin[13], Jochen Bauer[14], Francesco Benedetti[15,16], Klaus Berger[17], Bianca Besteher[18], Tiana Borgers[19], Robin Bülow[20], Li-Ping Cao[21], Guan-Mao Chen[22], Jian-Shan Chen[21], Si-Cheng Chen[2,23], Tao Chen[24], Tao-Lin Chen[25,26], Xiao Chen[2,4], Yan-Rong Chen[2,23], Yu-Qi Cheng[27], Zhao-Song Chu[28], Romain Colle[29,30], Colm G. Connolly[31], Emmanuelle Corruble[29,30], Baptiste Couvy-Duchesne[32,33,34,35], Xi-Long Cui[36], Kathryn R. Cullen[11,12], Udo Dannlowski[19,37,38,39], Christopher Davey[40], Danai Dima[41,42], Annemiek Dols[5,43], Tracy Erwin-Grabner[44], Jennifer W. Evans[45], Kira Flinkenflügel[19], Cynthia H. Y. Fu[46,47], Paola Fuentes-Claramonte[48,49], Qing-Lin Gao[2,4], Qi-Yong Gong[25,26], Ali Saffet Gönül[50], Ian H. Gotlib[51], Roberto Goya-Maldonado[44,52,53], Hans J. Grabe[54], Nynke Groenewold[55], Dominik Grotegerd[19], Wen-Bin Guo[36], Tim Hahn[19], Geoffrey B. Hall[56], J. Paul Hamilton[57], Ben Harrison[40], Sean Hatton[58], Can-Can He[59], Marco Hermesdorf[17], Ian B. Hickie[58], Tiffany C. Ho[60,61], Zheng-Jia-Yi Hu[2,62], Qian Huang[63], Neda Jahanshad[64], Hamidreza Jamalabadi[8], Alec Jamieson[40], Xin-Lei Ji[36], Feng-Nan Jia[65], Toshiharu Kamishikiryo[66], Tilo Kircher[8], Sheri-Michelle Koopowitz[55], Anna Kraus[19], Axel Krug[67], Li Kuang[63], Jim Lagopoulos[68], Thomas Lancaster[69,70,71], Nils Inge Landrø[9], Elisabeth J. Leehr[19], Meng Li[18], Bao-Juan Li[72], Feng Li[73], Hui-Xian Li[74], Tao Li[75,76], Xue-Ying Li[2,4], Yi-Fan Liao[2,4], Ze-Di Lin[77], David E. J. Linden[69,70,78], Hai-Long Liu[2,4], Xiao-Yun Liu[79], Zhe-Ning Liu[36], Yan-Song Liu[65], Yi-Cheng Long[36], Jian-Ping Lu[80], Bin Lu[2,4], Frank P. MacMaster[81,82,83], Luigi A. Maglanoc[84], Andrew M. McIntosh[85], Sarah Medland[86], David M. A. Mehler[69,70,87], Susanne Meinert[19,88], Benson Mwangi[89], Amar Ojha[90,91], Go Okada[66], Mardien L. Oudega[92,93], Brenda W. J. H. Penninx[5], Sara Poletti[15], Edith Pomarol-Clotet[48,49], Maria J. Portella[49,94,95], Jiang Qiu[96], Liesbeth Reneman[97], Henning S. Rise[98], Elena Rodriguez-Cano[48,49,99], Matthew D. Sacchet[100], Raymond Salvador[48,49], Hendrik Schlieper[19], Anouk Schrantee[97], Xiao-Xiao Shan[36], Hotaka Shinzato[66,101], Tian-Mei Si[102], Kang Sim[103,104,105], Jair C. Soares[89], Dan J. Stein[106], Benjamin Straube[8], Lachlan T. Strike[107], Peng-Feng Sun[108], Florian Thomas-Odenthal[8], Sophia I. Thomopoulos[64], Paul M. Thompson[64], Yara Toenders[109], Aslihan Uyar[50], Nic J. A. van der Wee[13], Steven J. A. van der Werff[13], Ilya M. Veer[110,111], Dick J. Veltman[5], Henry Völzke[112], Margot Wagenmakers[92,93], Henrik Walter[111], Martin Walter[18,113], Chuan-Yue Wang[73], Han-Lin Wang[2,62], Hua-Ning Wang[72], Xiang Wang[36], Xin-Yu Wang[2,23], Ying Wang[22], Yu-Wei Wang[114], Lars Tjelta Westlye[9,10,115], Heather C. Whalley[85], Sarah Whittle[116], Nils R. Winter[19], Katharina Wittfeld[54], Mon-Ju Wu[89], Xiao-Ping Wu[108], Xin-Ran Wu[96], Yan-Kun Wu[102], Chen-Nan Wu[2,4], Chun-Ming Xie[59], Guang-Rong Xie[36], Peng Xie[117], Xiu-Feng Xu[27], Zhen-Peng Xue[80], Tony T. Yang[118], Hong Yang[24], Jian Yang[119], Hua Yu[75,76], Yong-Qiang Yu[120,121], Min-Lan Yuan[122], Yong-Gui Yuan[79], Yu-Feng Zang[123,124], Carlos A. Zarate[45], Ai-Xia Zhang[125], Ke-Rang Zhang[125], Wei Zhang[122], Jing-Ping Zhao[36], Wen-Yi Zhong[62,126], Jia-Jia Zhu[120,121], Giovana B. Zunta-Soares[89], Xi-Nian Zuo[127,128], Elena Pozzi[129,130,131] & Lianne Schmaal[129,130,131] ✉

[1]Department of Psychological and Cognitive Sciences, Tsinghua University, Beijing, China. [2]State Key Laboratory of Cognitive Science and Mental Health, Institute of Psychology, Chinese Academy of Sciences, Beijing, China. [3]Research Center for Culture and Psychology, Tsinghua University, Beijing, China. [4]Department of Psychology, University of Chinese Academy of Sciences, Beijing, China. [5]Department of Psychiatry, Amsterdam UMC, Vrije Universiteit Amsterdam, Amsterdam, the Netherlands. [6]Amsterdam Neuroscience, Mood, Anxiety, Psychosis, Stress and Sleep Program, Amsterdam UMC, Vrije Universiteit Amsterdam, Amsterdam, the Netherlands. [7]Amsterdam Public Health, Mental Health Program, Amsterdam UMC, Vrije Universiteit Amsterdam, Amsterdam, the Netherlands. [8]Department of Psychiatry and Psychotherapy, University of Marburg, Marburg, Germany. [9]Clinical Neuroscience Research Group, Department of Psychology, University of Oslo, Oslo, Norway. [10]Centre for Precision Psychiatry, Oslo University Hospital and Institute of Clinical Medicine, University of Oslo, Oslo, Norway. [11]Department of Psychiatry and Behavioral Sciences, University of Minnesota Medical School, Minneapolis, MN, USA. [12]Masonic Institute of the Developing Brain, Minneapolis, MN, USA. [13]Department of Psychiatry, Leiden University Medical Center, Leiden, the Netherlands. [14]Department of Radiology, University of Münster, Münster, Germany. [15]Psychiatry and Clinical Psychobiology, Division of Neuroscience, IRCCS Ospedale San Raffaele, Milano, Italy. [16]University Vita-Salute San Raffaele, Milano, Italy. [17]Institute of Epidemiology and Social Medicine, University of Münster, Münster, Germany. [18]Department of Psychiatry and Psychotherapy, Jena University Hospital, Jena, Germany. [19]Institute for Translational Psychiatry, University of Münster, Münster, Germany. [20]Institute of Diagnostic Radiology and Neuroradiology, University Medicine Greifswald, Greifswald, Germany. [21]Affiliated Brain Hospital of Guangzhou Medical University, Guangzhou, China. [22]The First Affiliated Hospital of Jinan University, Guangzhou, China. [23]Magnetic Resonance Imaging Research Center, Institute of Psychology, Chinese Academy of Sciences, Beijing, China. [24]Department of Radiology, The First Affiliated Hospital, College of Medicine, Zhejiang University, Hangzhou, China. [25]Department of Radiology, Huaxi MR Research Center (HMRRC), Institute of Radiology and Medical Imaging, Psychoradiology Key Laboratory of Sichuan Province, West China Hospital of Sichuan University, Chengdu, China. [26]Xiamen Key Lab of Psychoradiology and Neuromodulation, Department of Radiology, West China Xiamen Hospital of Sichuan University, Xiamen, China. [27]Affiliated Mental Health Center and Hangzhou Seventh People's Hospital, Zhejiang University School of Medicine, Hangzhou, People's Republic of China. [28]Department of Psychiatry, First Affiliated Hospital of Kunming Medical University, Kunming, China. [29]MOODS Team, INSERM 1018, CESP, Faculté de Médecine Paris-Saclay, University of Paris-Saclay, Paris, France. [30]Service Hospitalo-Universitaire de Psychiatrie de Bicêtre, Groupe Hospitalier Paris-Saclay, Assistance Publique-Hôpitaux de Paris, Hôpital de Bicêtre, Le Kremlin Bicêtre, France. [31]Department of Biomedical Sciences, Florida State University, Tallahassee, FL, USA. [32]Sorbonne University, Paris Brain Institute (ICM), CNRS, Inria, Inserm, AP-HP, Hôpital de la Pitié Salpêtrière, Paris, France. [33]Institute for Molecular Bioscience, University of Queensland, St Lucia, Queensland, Australia. [34]Brain & Mental Health Program, QIMR Berghofer Medical Research Institute, Brisbane, Queensland, Australia. [35]School of Biomedical Sciences, Faculty of Health, Queensland University of Technology, Brisbane, Queensland, Australia. [36]Department of Psychiatry, and National Clinical Research Center for Mental Disorders, The Second Xiangya Hospital of Central South University, Changsha, China. [37]Department of Psychiatry, Medical School and University Medical Center OWL, Protestant Hospital of the Bethel Foundation, Bielefeld University, Bielefeld, Germany. [38]German Center for Mental Health (DZPG), Site Jena–Magdeburg–Halle, Jena–Magdeburg–Halle, Germany. [39]Center for Intervention and Research on Adaptive and Maladaptive Brain Circuits Underlying Mental Health (C-I-R-C), Site Jena–Magdeburg–Halle, Jena–Magdeburg–Halle, Germany. [40]Department of Psychiatry, University of Melbourne, Melbourne, Victoria, Australia. [41]Centre of Clinical, Social and Cognitive Neuroscience, Department of Psychology and Neuroscience, City St George's, University of London, London, UK. [42]Department of Neuroimaging, Institute of Psychiatry, Psychology and Neuroscience, King's College London, London, UK. [43]Department of Psychiatry, Division Brain, UMC Utrecht, Utrecht, the Netherlands. [44]Laboratory of Systems Neuroscience and Imaging in Psychiatry (SNIP-Lab), Department of Psychiatry and Psychotherapy, University Medical Center Göttingen (UMG), Göttingen, Germany. [45]Experimental Therapeutics and Pathophysiology Branch, National Institute of Mental Health, NIH, Bethesda, MD, USA. [46]Centre for Affective Disorders, School of Academic Psychiatry, Institute of Psychiatry, Psychology and Neuroscience, King's College London, London, UK. [47]Department of Psychology, University of East London, London, UK. [48]FIDMAG Germanes Hospitalaries Research Foundation, Barcelona, Spain. [49]Centro de Investigación Biomédica en Red de Salud Mental, Instituto de Salud Carlos III, Madrid, Spain. [50]Ege University School of Medicine SoCAT Lab Izmir, Izmir, Turkey. [51]Department of Psychology, Stanford University, Stanford, CA, USA. [52]Laboratory of Translational Psychiatry, Department of Psychiatry and Psychotherapy, Jena University Hospital (UKJ), Jena, Germany. [53]German Center for Mental Health (DZPG), partner site Jena, Jena, Germany. [54]Department of Psychiatry and Psychotherapy, University Medicine Greifswald, Greifswald, Germany. [55]Department of Psychiatry and Mental Health, Neuroscience Institute, University of Cape Town, Cape Town, South Africa. [56]Department of Psychology, Neuroscience and Behaviour, McMaster University, Hamilton, Ontario, Canada. [57]Department of Clinical and Biological Psychology, University of Bergen, Bergen, Norway. [58]Brain and Mind Centre, University of Sydney, Sydney, New South Wales, Australia. [59]Department of Neurology, Affiliated ZhongDa Hospital of Southeast University, Nanjing, China. [60]Department of Psychology, University of California, Los Angeles, CA, USA. [61]Department of Psychiatry and Behavioral Science, University of California, San Francisco, CA, USA. [62]Sino-Danish College, University of Chinese Academy of Sciences, Beijing, China. [63]Department of Psychiatry, The First Affiliated Hospital of Chongqing Medical University, Chongqing, China. [64]Imaging Genetics Center, Mark and Mary Stevens Neuroimaging and Informatics Institute, Keck School of Medicine of USC, University of Southern California, Marina del Rey, CA, USA. [65]Department of Clinical Psychology, Suzhou Psychiatric Hospital, The Affiliated Guangji Hospital of Soochow University, Suzhou, China. [66]Department of Psychiatry and Neurosciences, Graduate School of Biomedical and Health Sciences, Hiroshima University, Hiroshima, Japan. [67]Department of Psychiatry and Psychotherapy, University Hospital, Bonn, Germany. [68]Thompson Brain and Mind Healthcare, Maroochydore, Queensland, Australia. [69]Cardiff University Brain Research Imaging Centre, School of Psychology, Cardiff, UK. [70]MRC Centre for Neuropsychiatric Genetics and Genomics, School of Medicine, Cardiff University, Cardiff, UK. [71]Department of Psychology, University of Bath, Bath, UK. [72]Xijing Hospital of Air Force Military Medical University, Xi'an, China. [73]Beijing Anding Hospital, Capital Medical University, Beijing, China. [74]The Third Affiliated Hospital of Zhengzhou University, Zhengzhou, China. [75]Affiliated Mental Health Center and Hangzhou Seventh People's Hospital, Zhejiang University School of Medicine, Hangzhou, China. [76]Mental Health Center and Psychiatric Laboratory, West China Hospital of Sichuan University, Chengdu, China. [77]Beijing Key Laboratory of Learning and Cognition, School of Psychology, Capital Normal University, Beijing, China. [78]Department of Psychiatry and Neuropsychology, Mental Health and Neuroscience Institute, Maastricht University, Maastricht, the Netherlands. [79]Department of Psychosomatics and Psychiatry, Zhongda Hospital, School of Medicine, Southeast University, Nanjing, China. [80]Shenzhen Kangning Hospital, Shenzhen, China. [81]University of Calgary, Calgary, Alberta, Canada. [82]Department of Psychiatry, Dalhousie University, Halifax, Nova Scotia, Canada. [83]IWK Health, Halifax, Nova Scotia, Canada. [84]Data Management, IT Department, University of Oslo, Oslo, Norway. [85]Division of Psychiatry, Centre for Clinical Brain Science, University of Edinburgh, Edinburgh, UK. [86]Psychiatric Genetics, QIMR Berghofer, Brisbane, Queensland, Australia. [87]Department of Psychiatry, Psychotherapy and Psychosomatics, Medical School, Uniklinik RWTH Aachen, Aachen, Germany. [88]Institute for Translational Neuroscience, University of Münster, Münster, Germany. [89]Center of Excellence on Mood Disorders, Louis A. Faillace, MD, Department of Psychiatry and Behavioral Sciences, The University of Texas Health Science Center at Houston, Houston, TX, USA. [90]Center for Neuroscience, University of Pittsburgh, Pittsburgh, PA, USA. [91]Center for the Neural Basis of Cognition, University of Pittsburgh, Pittsburgh, PA, USA. [92]Department of Psychiatry, Boelelaan 1117, Amsterdam UMC, Vrije Universiteit Amsterdam, Amsterdam, the Netherlands. [93]GGZ inGeest Mental Health Care, De Nieuwe Valerius, Amsterdam, the Netherlands. [94]Sant Pau Mental Health Research Group, Institut de Recerca Sant

Pau, Barcelona, Spain. [95]Department of Psychiatry and Forensic Medicine, Universitat Autònoma de Barcelona, Bellaterra, Spain. [96]Faculty of Psychology, Southwest University, Chongqing, China. [97]Amsterdam UMC, Department of Radiology and Nuclear Medicine, University of Amsterdam, Amsterdam, the Netherlands. [98]Department of Psychology, University of Oslo, Oslo, Norway. [99]Consorci Sanitari del Maresme, Mataró, Spain. [100]Meditation Research Program, Department of Psychiatry, Massachusetts General Hospital, Harvard Medical School, Boston, MA, USA. [101]Department of Neuropsychiatry, Graduate School of Medicine, University of the Ryukyus, Okinawa, Japan. [102]National Clinical Research Center for Mental Disorders (Peking University Sixth Hospital) and Key Laboratory of Mental Health, Ministry of Health (Peking University), Beijing, China. [103]West Region, Institute of Mental Health, Singapore, Singapore. [104]Yong Loo Lin School of Medicine, National University of Singapore, Singapore, Singapore. [105]Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore, Singapore. [106]SAMRC Unit on Risk and Resilience in Mental Disorders, Department of Psychiatry and Mental Health, University of Cape Town, Cape Town, South Africa. [107]Brain and Mental Health, QIMR Berghofer, Brisbane, Queensland, Australia. [108]Xi'an Central Hospital, Xi'an, China. [109]Erasmus School of Social and Behavioral Sciences, Erasmus University, Rotterdam, the Netherlands. [110]Department of Developmental Psychology, Psychology Research Institute, University of Amsterdam, Amsterdam, the Netherlands. [111]Department of Psychiatry and Psychotherapy, Charité–Universitätsmedizin Berlin (corporate member of FU Berlin and Humboldt Universität zu Berlin), Berlin, Germany. [112]Institute for Community Medicine, University Medicine Greifswald, Greifswald, Germany. [113]German Center for Mental Health (DZPG), German Center for Mental Health (partner site Halle–Jena), Magdeburg, Germany. [114]Department of Psychiatry, Yale School of Medicine, New Haven, CT, USA. [115]KG Jebsen Centre for Neurodevelopmental Disorders, University of Oslo, Oslo, Norway. [116]School of Psychology, Deakin University, Burwood, Victoria, Australia. [117]Institute of Neuroscience, Chongqing Medical University, Chongqing, China. [118]Department of Psychiatry and Behavioral Sciences, Division of Child and Adolescent Psychiatry, Weill Institute for Neurosciences, University of California, San Francisco (UCSF), San Francisco, CA, USA. [119]Chongqing Key Laboratory of Neurobiology, Chongqing, China. [120]Department of Radiology, The First Affiliated Hospital of Anhui Medical University, Hefei, China. [121]Research Center of Clinical Medical Imaging, Hefei, China. [122]West China Hospital of Sichuan University, Chengdu, China. [123]Precision TMS Center/Department of Neurology, The Affiliated Hospital of Hangzhou Normal University, Hangzhou, China. [124]Zhejiang Key Laboratory for Research in Assessment of Cognitive Impairments, Hangzhou, China. [125]First Hospital of Shanxi Medical University, Taiyuan, China. [126]School of Biomedical Engineering, Southern Medical University, Guangzhou, China. [127]Developmental Population Neuroscience Research Center, IDG/McGovern Institute for Brain Research, Beijing Normal University, Beijing, China. [128]National Basic Science Data Center, Beijing, China. [129]Centre for Youth Mental Health, University of Melbourne, Melbourne, Victoria, Australia. [130]Orygen, Parkville, Victoria, Australia. [131]These authors contributed equally: Elena Pozzi, Lianne Schmaal. ✉e-mail: yancg@tsinghua.edu.cn; lianne.schmaal@unimelb.edu.au

nature portfolio

Corresponding author(s): Chao-Gan Yan and Lianne Schmaal

Last updated by author(s): 3/2/2026

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's d, Pearson's r), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Structural T1-weighted MRI scans and clinical data were collected at 64 cohorts worldwide participating in the DIRECT (Chinese) and ENIGMA MDD (international) consortia. Each site acquired data using local protocols approved by its institutional ethics committee and contributed de-identified demographic, clinical and MRI measures.
Data analysis	Cortical reconstruction and surface-based morphometry were performed with FreeSurfer following harmonized DIRECT and ENIGMA protocols. Vertex-wise cortical thickness and surface area maps were analyzed using linear regression models within each cohort. Within-site t values were converted into Cohen's d and entered into a random-effect meta-model (using R metansue, https://www.metansue.com/) for voxel-wise meta-analysis (integrated as y_Meta_Image_CallR.m in DPABI). All custom analysis scripts have been made publicly available at https://github.com/Chaogan-Yan/PaperScripts/tree/master/Yan_2025.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The dataset required to interpret, replicate and extend the findings of this study consists of the derived meta-analytic statistical maps and summary data generated during the analyses for all participating sites. These data are available at https://doi.org/10.57760/sciencedb.29686. Individual-level participant data from the contributing cohorts cannot be publicly shared due to local ethics and data-sharing restrictions across participating sites.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Sex was recorded as binary (male/female) at each site based on self-report or clinical records. Additional models tested diagnosis-by-sex interactions on cortical measures. Sex distributions for each cohort are reported in the Supplementary Tables.
Reporting on race, ethnicity, or other socially relevant groupings	Race/ethnicity was not available at the same level of detail across all cohorts. As a proxy for broad population background, we distinguished DIRECT cohorts (Chinese participants) from ENIGMA MDD cohorts (predominantly Caucasian with some other backgrounds) and performed meta-regression on the proportiaon of Caucasian, Asian and African participants to explore potential population-level moderators, as reported in the Supplement.
Population characteristics	The study included 12,274 participants (5,736 patients with major depressive disorder and 6,538 healthy controls) from 64 cohorts worldwide. Samples covered adolescents and adults, with variation in illness stage (first-episode, recurrent, remitted), medication status, and age of onset; detailed cohort-level demographics and clinical characteristics are provided in the Supplementary Tables.
Recruitment	Participants with MDD were recruited from psychiatric clinics and hospitals at each site, and healthy controls from the local community via advertisements, volunteer panels or registries. Diagnostic status was established using structured clinical interviews or clinician-based diagnoses following standard criteria (e.g., DSM or ICD), as summarized for each cohort in the Supplementary material.
Ethics oversight	All cohorts obtained approval from their local institutional review boards or ethics committees, and all participants (or their legal guardians for minors) provided written informed consent. Data shared within the DIRECT and ENIGMA MDD consortia were de-identified before transfer to the central analysis sites. For the DIRECT datasets, approval was obtained from the Ethics Committee of the Institute of Psychology, Chinese Academy of Sciences. For ENIGMA MDD, ethics approval for the storage and management of consortium data was granted by the University of Melbourne Human Research Ethics Office. All participating sites obtained approval from their respective local institutional review boards or ethics committees, and all participants provided written informed consent.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	All available participants with usable structural MRI data and clinical information from 64 cohorts were included (12,274 individuals: 5,736 MDD and 6,538 controls). No prospective power calculation was performed because this is a large multi-cohort consortium study, but the sample size substantially exceeds previous structural MRI studies of MDD.
Data exclusions	Participants were excluded if MRI scans failed local or central quality control (e.g., motion or segmentation failure) or if key demographic/clinical data were missing ("Sample Selection"). Cohort-level inclusion and exclusion criteria and the numbers excluded at each step are summarized in the Supplementary Tables.
Replication	Reproducibility was supported by harmonized image-processing and statistical-analysis pipelines, standardized quality-control procedures, and site-level analyses combined through meta-analysis across cohorts.

Randomization	This is an observational case–control study; participants were not randomized. Group membership (MDD vs control) was determined by clinical diagnosis at each site.
Blinding	MRI data were processed and analyzed using de-identified datasets. Clinical raters establishing diagnoses were independent from imaging analysts and were unaware of the central meta-analytic results.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☐	☒ MRI-based neuroimaging

Clinical data

Policy information about clinical studies
All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	Not applicable; this study is not a clinical trial.
Study protocol	Not applicable; imaging and clinical data were collected as part of observational cohort studies at each site under local ethics approvals.
Data collection	Clinical assessments (e.g., diagnostic interviews, symptom ratings, age of onset, episode status, medication use) and structural MRI scans were collected at participating sites over an extended period as part of routine clinical or research protocols.
Outcomes	The primary outcomes were vertex-wise differences in cortical thickness and surface area between patients with major depressive disorder and healthy controls. Secondary outcomes included associations with age, sex, age of onset, illness stage (first-episode vs recurrent), symptom severity and medication status, as described in the Methods.

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*

Magnetic resonance imaging

Experimental design

Design type	Cross-sectional multi-cohort case–control study using structural MRI.
Design specifications	Single 3D T1-weighted structural scan per participant acquired at rest. Analyses compared cortical thickness and surface area between MDD and control groups and examined moderators such as age, sex, age of onset, illness stage and medication status.

Behavioral performance measures	No behavioral task was performed during MRI acquisition; only clinical and demographic measures were collected.

Acquisition

Imaging type(s)	Structural T1-weighted MRI; no functional or diffusion MRI were analyzed.
Field strength	Data were acquired on 1.5T and 3T scanners across cohorts; field strength for each site is listed in the Supplementary Tables.
Sequence & imaging parameters	Sites used their local 3D T1-weighted sequences. Key acquisition parameters for each scanner and sequence are summarized in the Supplementary Tables.
Area of acquisition	Whole brain

Diffusion MRI ☐ Used ☒ Not used

Preprocessing

Preprocessing software	Cortical reconstruction and parcellation were performed with FreeSurfer (versions 5.x or 6.x depending on cohort) following harmonized DIRECT and ENIGMA MDD pipelines, implemented via DPABISurf where applicable
Normalization	Individual cortical surfaces were registered to the FreeSurfer fsaverage template for vertex-wise analyses.
Normalization template	FreeSurfer fsaverage surface.
Noise and artifact removal	Each site applied visual quality control of raw T1-weighted images and FreeSurfer segmentations; participants with motion artefacts, major segmentation errors or other artefacts were excluded.
Volume censoring	Not relevant.

Statistical modeling & inference

Model type and settings	Within each cohort, vertex-wise general linear models were used to estimate the effect of diagnosis (MDD vs control) on cortical thickness and surface area. Cohort-specific effect sizes (Cohen's d) and standard errors were then combined using meta-analyses.
Effect(s) tested	Primary effects tested were group differences (MDD vs control) in cortical thickness and surface area, as well as diagnosis-by-age and diagnosis-by-sex interactions and associations with illness characteristics (age of onset, episode status, remission, medication). Analyses were primarily whole-brain vertex-wise; summary statistics were also derived for clusters and regions for visualization.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☒ Both

Anatomical location(s)	Anatomical labels in the figures were derived from the Desikan–Killiany cortical atlas for visualization only.
Statistic type for inference (See Eklund et al. 2016)	Inference was based on vertex-wise test statistics (t-values and corresponding Cohen's d effect sizes).
Correction	Vertex-wise P values were corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate procedure ($q < 0.05$), applied separately for cortical thickness and surface area across all cortical vertices in the fsaverage space.

Models & analysis

n/a	Involved in the study
☒	☐ Functional and/or effective connectivity
☒	☐ Graph analysis
☒	☐ Multivariate modeling or predictive analysis